

Programming Technology Documentation

2nd assignment task 6

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Task

Four game

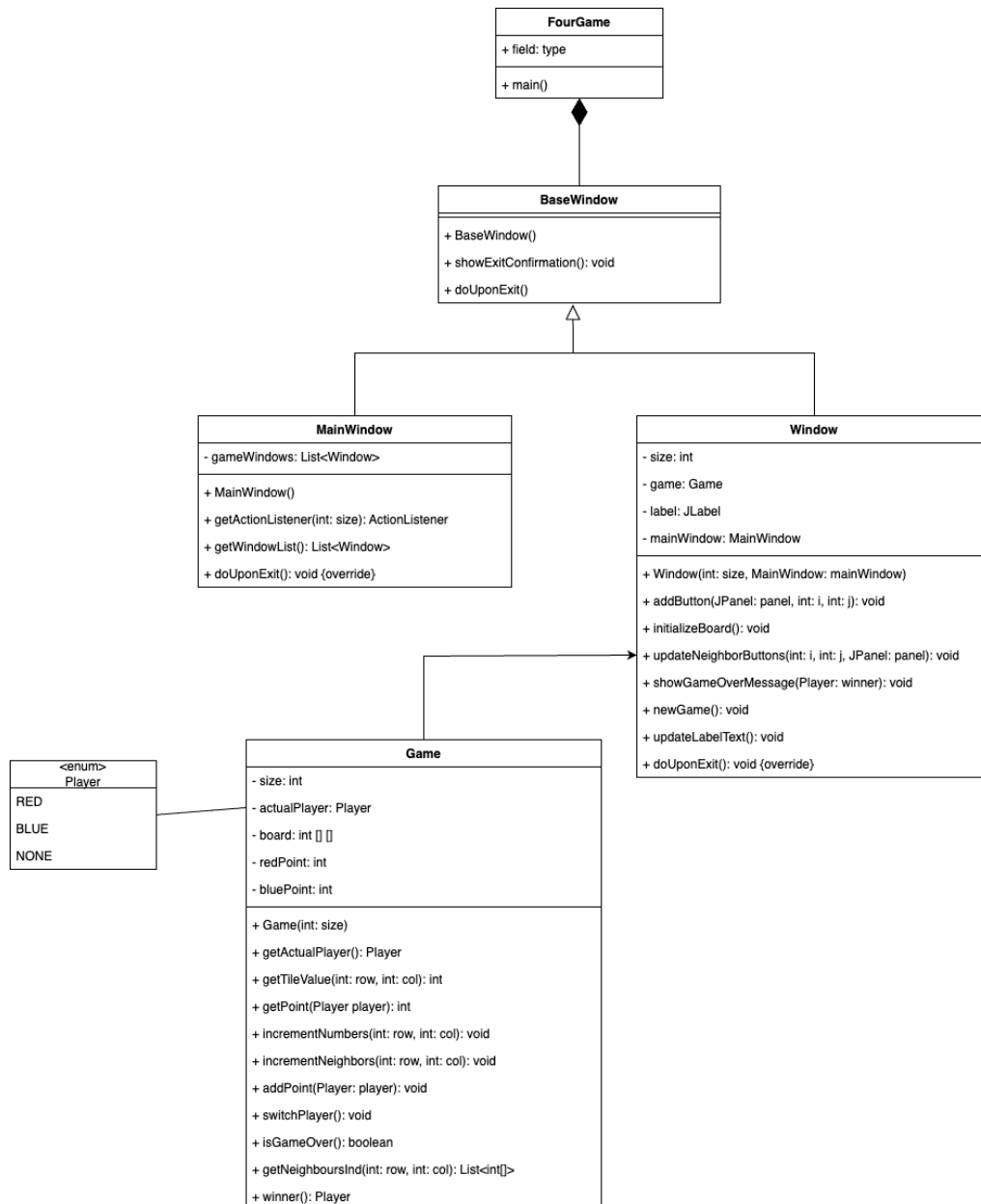
This a two-player game is played on a board consists of $n \times n$ fields, where each field contains a value between 0 and 4. Initially, all the fields contain the value of 0. If a player chooses a field, then the value of the field and its neighbours incremented by one (if the value is less than 4). The player's score represents how many fields did he make to have the value of 4. If a value of a field reaches 4, then the field is colorized with the color of the actual player (red or blue). The game ends, when all fields have the value of 4. The player having the higher score wins.

Implement this game, and let the board size be selectable (3x3, 5x5, 7x7). The game should recognize if it is ended, and it has to show in a message box which player won. After this, a new game should be started automatically.

Task Analysis

- When a player choose a field, that field and its neighboring fields' values are incremented (max is 4).
- If any neighbour field's value reaches 4, it will change color to the current player's color (red or blue).
- The game ends, when all fields have the value of 4.
- The player having the higher score wins.
- At the end of the game, a message box will display the winner.
- After the game ends, a new game is started automatically with the same board size.

Class diagram



Method Descriptions

[BaseWindow.java]

showExitConfirmation(): Ask if the user really wants to close the window or not (if the user clicked the closing button).

doUponExit(): Close the window.

[MainWindow.java]

getActionListener(int size): Assign an appropriate event listener to each button or menu item to start a game of the specified size.

[Window.java]

Window(int size, MainWindow mainWindow): Make a game window.

addButton(JPanel panel, int i, int j): Make a tile and set its function when it's clicked.

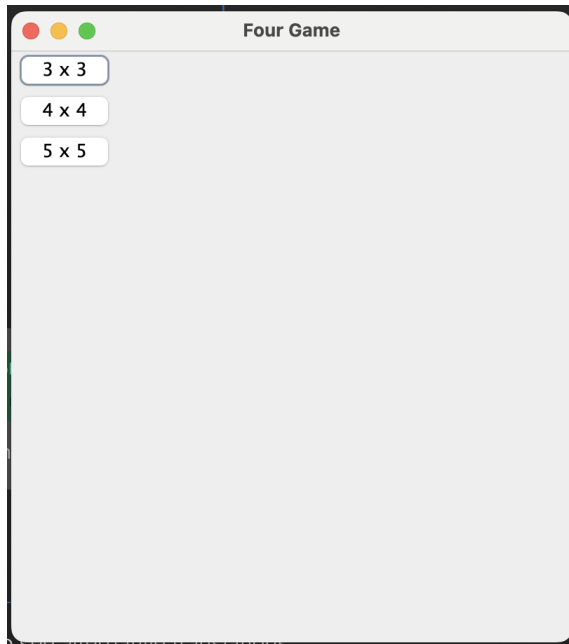
updateNeighborButtons(int i, int j, JPanel panel): Update the tile's display value.

showGameOverMessage(Player winner): Display who won (can be Draw in 4x4).

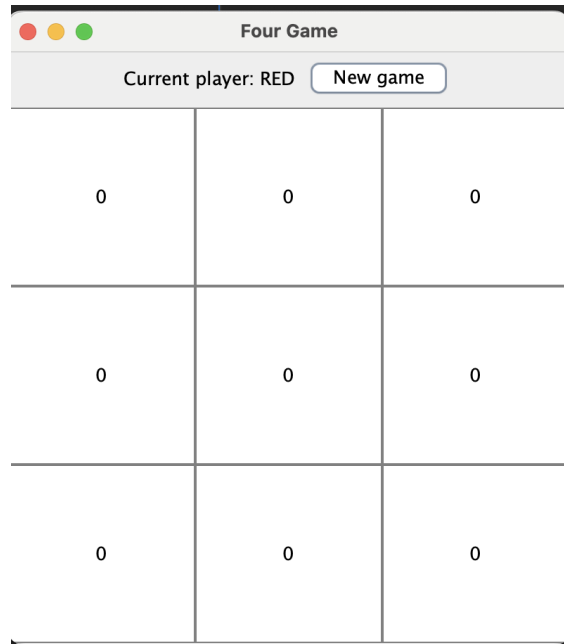
newGame(): Start a new game when the game is finished.

Application Screenshots

Home screen:



3x3:



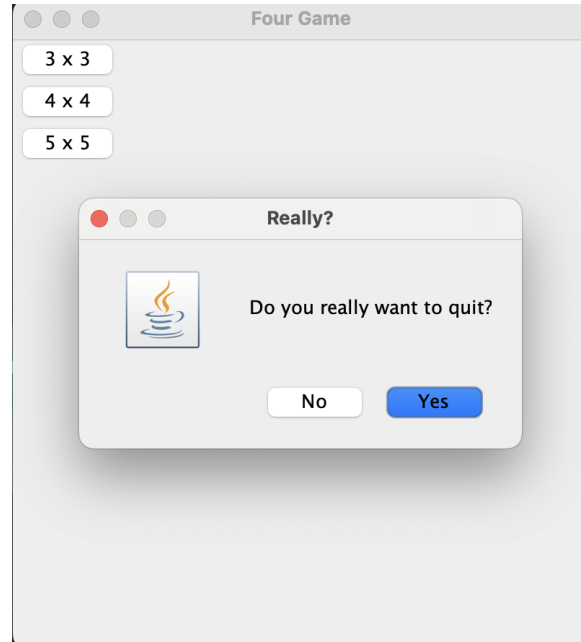
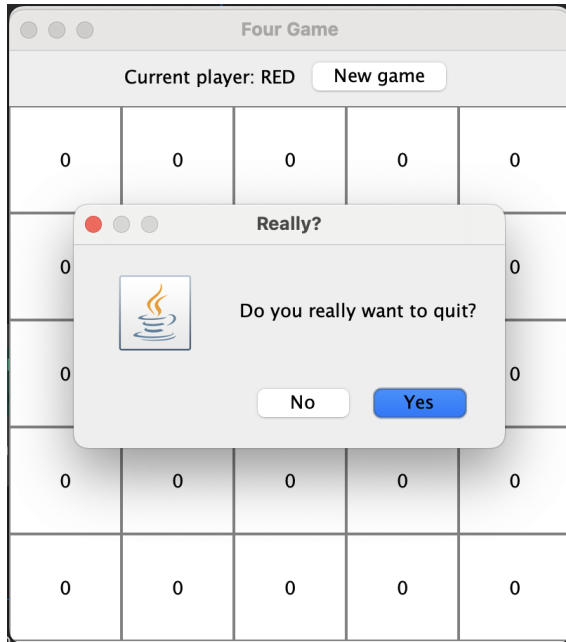
4x4:



5x5:



Closing screen:



Test Cases

Test Case 1: Display the main menu

AS A user

I WANT TO see the main menu options

GIVEN the application is launched

WHEN the main window is displayed

THEN I should see buttons for "3 x 3", "4 x 4", and "5 x 5".

Test Case 2: Start a 3x3 game

AS A user

I WANT TO start a 3x3 game

GIVEN the main menu is visible

WHEN I click the "3 x 3" button

THEN a new window with a 3x3 grid should appear.

Test Case 3: Ensure tiles are initialized

AS A user

I WANT TO see all tiles initialized to 0

GIVEN a new game has started

WHEN I look at the game board

THEN all tiles should display "0".

Test Case 4: Increment tile value

AS A user

I WANT TO increase the value of a tile

GIVEN a game board is active

WHEN I click on a tile with a value less than 4

THEN the tile's value should increment by 1.

Test Case 5: Neighbor increment logic

AS A user

I WANT TO see neighboring tiles increment

GIVEN a game board is active

WHEN I click on a tile with a value less than 4

THEN the neighbours tiles (above, under, left, right) should increment by 1, if their value is less than 4.

Test Case 6: Prevent increment on full tile

AS A user

I WANT TO prevent incrementing a full tile

GIVEN a tile has a value of 4

WHEN I click on it

THEN the value of the tile should remain 4.

Test Case 7: Change tile color to player's color

AS A user

I WANT TO change the tile color when it reaches 4

GIVEN a tile's value is 4

WHEN the tile value changes from 3 to 4

THEN the tile color should change to the current player's color.

Test Case 8: Prevent changing a tile with value 4

AS A user

I WANT TO avoid modifying the tiles with a tile with value 4

GIVEN a tile's value is 4 and has a color

WHEN I click on it

THEN no changes should occur to the tile or its neighbors.

Test Case 9: Player switch after turn

AS A user

I WANT TO switch player turns

GIVEN it is Player RED's turn

WHEN they make a move

THEN the next turn should belong to Player BLUE.

Test Case 10: Check game over condition

AS A user

I WANT TO end the game when all tiles are full

GIVEN the game board is active

WHEN all tiles reach the value of 4

THEN the game should display the winner or a draw message.

Test Case 11: Calculate player points

AS A developer

I WANT TO verify the score calculation

GIVEN some tiles are colorized by each player

WHEN the game calculates points

THEN the total points for each player should equal the number of tiles they own.

Test Case 12: Display winner

AS A user

I WANT TO see the game winner

GIVEN the game ends

WHEN one player has more points than the other

THEN the game should display the winner name.

Test Case 13: Restart game after completion

AS A user

I WANT TO start a new game after the current one ends

GIVEN the game ends

WHEN I confirm the end message

THEN a new game should start.

Test Case 14: New game from menu

AS A user

I WANT TO start a new game

GIVEN I am playing a game

WHEN I click the "New game" button

THEN the current game should close, and a new one should start.

Test Case 15: Exit game confirmation

AS A user

I WANT TO confirm before exiting

GIVEN I click the close button

WHEN the exit confirmation dialog appears

THEN the application should only close if I select "Yes".